

From Field to Fork: A Data-Driven Framework for Advancing Regenerative Farming and Food Safety

Pesticide Usage and Application: A Historical and Contemporary Record

A comprehensive understanding of pesticide usage in the United States, stretching back nearly a century, is not found within a single repository but rather through the aggregation of multiple federal programs and state-level reporting systems. These sources collectively provide a detailed picture of application volumes, chemical composition shifts, and geographic distribution, forming a critical pillar for any advocacy effort aimed at regulatory change. The primary federal entities responsible for tracking these metrics are the U.S. Geological Survey (USGS), the U.S. Department of Agriculture (USDA), and the Environmental Protection Agency (EPA). While robust national records begin in the latter half of the 20th century, they offer invaluable longitudinal data for analyzing trends.

The most significant resource for historical trends in agricultural pesticide use is the USGS's Pesticide National Synthesis Project (PNSP) ¹²⁹. This project was created to consolidate and analyze farm survey data to produce county-level estimates of annual agricultural pesticide use dating back to the 1970s ¹³⁰¹³¹. The PNSP provides spatial data layers in GIS format and offers downloadable tables and maps for more than 500 different pesticides, allowing for detailed geographical and temporal analysis across the conterminous United States ¹³²¹³³. These data have been instrumental in assessing water quality impacts and developing benchmarks for pesticide concentrations in streams ⁹⁰. Although the project faced periods of suspension and restoration, its core function remains a cornerstone for demonstrating long-term increases in chemical inputs and shifts in pesticide chemistries ¹²⁹. For instance, the PNSP has provided data extending back to the 1970s for many chemicals, offering a crucial baseline before the widespread adoption of newer classes of pesticides ¹²⁸.

Building upon this foundation, several USDA programs provide official statistics and residue monitoring data. The USDA's Agricultural Marketing Service (AMS) has conducted a program to collect data on pesticide residues in food since the mid-20th century, with some historical data available as far back as the 1940s ^{44 79}. A report compiled by the USDA itself details trends in pesticide quantities used during the 20th century, providing context from 1931 through the 1960s ⁴¹. More formally, the USDA's Agricultural Chemical Use Program (ACUP), managed by the National Agricultural Statistics Service (NASS), began collecting and publishing official statistics on on-farm chemical use in the early 1990s ¹¹⁰¹¹⁶. This program was partly initiated in response to public concerns about pesticide exposure, such as the Alar scare in the early 1990s ¹¹⁶. Concurrently, the USDA's Pesticide Data Program (PDP), also under AMS, serves as a national monitoring program that analyzes food commodities for pesticide residues ¹¹¹. With over 32 years of available data, the PDP represents one of the largest sources of food pesticide residue data in the world, generating close to 42 million data points over three decades ^{43 112}. This program is essential for FDA risk assessments, as it informs evaluations of dietary exposure and the establishment of tolerances for pesticide residues allowed on food ⁹⁸.

State-level initiatives, particularly California's, offer a level of granularity unmatched by federal programs. The California Department of Pesticide Regulation (CDPR) operates the California Pesticide Information Portal (CalPIP), a powerful tool for accessing searchable, granular pesticide use data ⁴⁹. CalPIP allows users to query data starting from 1990, broken down by calendar year, county, ZIP code, crop, and even specific product names ^{81 147151}. This enables hyper-local analysis and the identification of specific regions or agricultural sectors with high pesticide application rates ¹⁴⁵. The portal aggregates data from multiple sources, including the Pesticide Use Reporting (PUR) system, which contains detailed records of all pesticide applications reported in the state ¹⁴⁸¹⁴⁹. This wealth of data, available for download, makes California a model for what is possible when states prioritize transparency in pesticide reporting ¹⁴⁶.

While comprehensive century-long datasets are unavailable, a synthesis of these sources reveals a clear trajectory. Early records from the 1940s show initial data collection efforts focused on residues ⁷⁹. The PNSP provides the first large-scale, county-level view from the 1970s onward ¹²⁹. The ACUP and PDP solidified official federal statistics and monitoring in the 1990s ¹¹⁰¹¹¹. State-level portals like CalPIP offer highly detailed data from 1990 forward ¹⁵¹. This multi-layered record is sufficient to document the dramatic increase in total pesticide volume applied to U.S. agriculture, the shift in chemical classes from organochlorines to organophosphates, carbamates, pyrethroids, neonicotinoids, and

glyphosate, and the increasing prevalence of seed treatments ¹⁰²¹³⁷. An advocacy report can effectively leverage these sources to map these changes geographically and temporally, highlighting areas where specific, well-studied pesticides have become dominant.

Data Source	Primary Focus	Time Period Covered	Geographic Scope	Key Features
Pesticide National Synthesis Project (USGS)	County-level agricultural pesticide use estimation	1970s - Present	Conterminous U.S.	Aggregates farm survey data; maps & tables for >500 pesticides; shows spatial/temporal trends ¹²⁹¹³¹ .
Agricultural Chemical Use Program (USDA NASS)	Official on-farm chemical use statistics	Early 1990s - Present	U.S.	Federal source for pest management practices; partial response to 1990s public concern ¹¹⁰¹¹⁶ .
Pesticide Data Program (USDA AMS)	Monitoring of pesticide residues on food	~1992 - Present	U.S.	Largest food residue database; 32+ years of data; informs FDA tolerance levels ^{43 98 111} .
California Pesticide Information Portal (CalPIP)	Granular, searchable pesticide use reporting	1990 - Present	California	Hyper-local data by county, ZIP code, crop; searchable by chemical name and product ^{49 81 145} .
USDA History Collection	Archival reports and data	1930s - 1990s	U.S.	Contains early land use planning reports and historical agricultural data ⁴² .

This array of data sources provides a robust, albeit fragmented, historical record. By triangulating information from the USGS PNSP, USDA ACUP/PDP, and state-level portals like CalPIP, a compelling narrative can be constructed to demonstrate the scale and evolution of pesticide use in the United States, directly supporting the need for regulatory review and potential bans on certain substances.

Soil Health and Regenerative Agriculture: Long-Term Experimental and Survey Data

The concept of "regenerative farming" seeks to reverse soil degradation and enhance ecosystem services, yet quantifying its long-term benefits requires access to decades-spanning datasets. While no single source tracks the adoption of regenerative practices over a full century, two major federally supported networks—the Long-Term Agroecosystem Research (LTAR) Network and the Long-Term Ecological Research (LTER) Network—provide invaluable, long-term experimental data on soil health indicators. Complementing these are statistical surveys like the National Resources

Inventory (NRI), which offer a broader, though less intensive, perspective on national soil trends.

The cornerstone of modern soil health research is the USDA's Long-Term Agroecosystem Research (LTAR) network ³². This network comprises 18 to 19 established research sites distributed across diverse agroecosystems in the contiguous United States, each with an average history of over 50 years ^{48 121}. The LTAR network was designed to understand and enhance the sustainability of American agriculture through coordinated, cross-site research ^{34 38}. It systematically combines strategic research projects with common measurements across its sites, producing a vast and growing body of data on soils, water, biodiversity, and ecosystem services ^{69 87}. Crucially for this research goal, LTAR data includes key indicators of soil health such as soil organic carbon (SOC), erosion rates, and nutrient cycling ^{87 138}. One study highlighted that LTAR-led research contributed to a 20% increase in SOC at a depth of 0-12 inches under specific management systems, providing empirical evidence of the benefits sought by regenerative advocates ⁷⁶. The data from these long-term experiments is being centralized in the LTAR Open Data Hub, although discovery tools are still evolving ⁴⁶. Working groups within LTAR are actively developing a common language and web-based tools to inventory and make accessible soils data from all 19 sites, facilitating comparative analyses between conventional and conservation-focused systems ^{88 136}.

Parallel to the LTAR network is the Long-Term Ecological Research (LTER) Network, funded by the National Science Foundation (NSF) ¹²⁶. This network consists of 28 sites dedicated to conducting ecological research over extended periods ¹²⁶. The Kellogg Biological Station (KBS) LTER, established in 1989, is a prominent example focused exclusively on understanding the ecology of agricultural landscapes ^{123 125}. KBS LTER research examines the complex interactions between soils, plants, microbes, and insects, providing fundamental insights into how agricultural practices shape agroecosystems ¹²³. The network's commitment to archiving samples and data ensures that future researchers can conduct retrospective analyses, making it a valuable resource for tracking long-term change ^{51 124}. Both the LTAR and LTER networks represent a gold standard for collecting data on the very processes that define regenerative versus non-regenerative systems.

Beyond these intensive research networks, the Natural Resources Conservation Service (NRCS) conducts the National Resources Inventory (NRI), a statistical survey of natural resources on non-Federal land in the United States ¹²⁷. The NRI's collection of data began in the 1930s, driven by the crisis of the Dust Bowl, making it the oldest continuous source of nationwide soil erosion data ^{54 57}. The NRI captures data on land cover, soil erosion,

wetlands, and conservation practices, providing a broad overview of resource conditions and trends ⁵⁵. A USDA-NRCS report based on NRI data showed that soil erosion rates on cropland decreased by 35% between 1982 and 2017, a testament to the effectiveness of decades of conservation policy ⁵⁶. While the NRI measures outcomes rather than the specific farming practices causing them, this data is essential for framing the problem of soil degradation and demonstrating the partial success of current conservation efforts. This creates a powerful narrative arc: acknowledging past problems and modest improvements sets the stage for arguing that more advanced, regenerative approaches are needed to achieve true sustainability.

In summary, while direct, century-long monitoring of "regenerative" practices is rare, the combination of LTAR and LTER provides decades of high-quality, controlled experimental data on soil health. This can be supplemented by the long-term trend data from the NRI. Together, these sources allow for a quantitative assessment of the impacts of different management systems on soil organic carbon, erosion, and overall ecosystem function, providing the empirical backbone for an advocacy campaign promoting regenerative agriculture.

Crop Nutritional Composition: Tracing Trends and Contested Narratives

The claim that the nutritional content of our food supply has declined over the past century is a central theme in arguments for agricultural reform. The primary source for investigating this question is the U.S. Department of Agriculture's (USDA) food composition database, a resource that has tracked the nutrient content of foods for over 125 years ²⁹. However, the interpretation of this data is complex and contested, requiring a nuanced approach for any advocacy effort.

The most widely cited evidence for nutritional decline comes from a 2004 study published in the *Journal of the American College of Nutrition*, which analyzed USDA data and concluded that the concentration of key nutrients like protein, calcium, iron, potassium, and vitamin C in 43 garden crops had decreased significantly between 1950 and 1999 ²⁹. This finding, derived from the USDA's National Nutrient Database for Standard Reference (SR), resonated widely and became a cornerstone of the "nutritionally depleted food" argument ^{25 28}. The SR database, now integrated into the more modern FoodData Central platform, contains historical data derived from analyses,

calculations, and published literature, providing a long-term record of macronutrients, micronutrients, and other compounds ^{24 28}. Other analyses using SR data have also pointed towards declines; for example, a study on city-pair analytical composites created from samples collected in 12 U.S. cities between 1940 and 1958 provided foundational data for nutrient comparisons ¹⁵⁹.

However, this narrative of universal decline is not universally accepted within the scientific community. A subsequent analysis of the same USDA SR database, covering the period from 1950 to 1999, found no statistically significant decline in the concentrations of eight key minerals: calcium, copper, iron, magnesium, manganese, potassium, sodium, and zinc ¹⁵⁸. This discrepancy highlights the profound impact of methodology on conclusions. Factors such as changes in crop cultivars bred for higher yields or transportability rather than nutrient density, variations in climate and weather patterns, differing soil mineral availability (as documented by the NRI and LTAR networks), and differences in analytical techniques over time can all confound simple time-series analyses of nutrient content ²⁹. Therefore, presenting a blanket statement that all nutrients are declining is scientifically unsound and risks undermining the credibility of an advocacy report if challenged.

A more robust and defensible strategy is to reframe the argument away from a simplistic narrative of universal decline and toward the concept of **nutritional instability**. This framing acknowledges the debate while focusing on the negative implications of industrial agricultural practices. The argument could posit that the modern food system, characterized by monocultures, heavy reliance on synthetic fertilizers, and widespread pesticide use, may be prioritizing biomass production over the development of dense, resilient plant biochemistry. Synthetic nitrogen fertilizers, for instance, can promote rapid growth (increasing yield) without a corresponding increase in the uptake of micronutrients from the soil, potentially leading to crops with lower nutritional quality. This hypothesis can be linked directly to the soil health data from the LTAR network, which tracks soil organic matter and microbial communities essential for nutrient cycling ¹³⁶¹³⁹. By connecting the dots between soil degradation, fertilizer inputs, and fluctuating crop nutrient profiles, a more compelling causal argument can be made. The goal would be to argue not that every apple today has less vitamin C than an apple in 1950, but that the food supply as a whole is becoming less predictable in its nutritional profile, making it harder for individuals to obtain optimal nutrition from diet alone. This nuance strengthens the scientific basis of the advocacy effort and prepares it to withstand scrutiny.

Toxicology and Human Exposure: Linking Chemicals to Health Outcomes

The most direct evidence linking agricultural practices to public health is found in the fields of toxicology and human biomonitoring. This body of work provides the scientific foundation for arguing that the presence of pesticides in the environment and on our food leads to measurable harm. The primary sources for this evidence are the National Toxicology Program (NTP), the EPA's Integrated Risk Information System (IRIS), and the Centers for Disease Control and Prevention's (CDC) National Health and Nutrition Examination Survey (NHANES).

The National Toxicology Program (NTP), part of the National Institutes of Health (NIH), is a key interagency group responsible for evaluating the toxicity of substances of public health concern ²¹. The NTP conducts rigorous, long-term animal studies to characterize the carcinogenic activity, genetic toxicity, and other harmful effects of thousands of chemicals, including numerous pesticides ^{20 21}. The program has produced extensive data on agents like glyphosate, atrazine, paraquat, and 2,4-D, examining their potential to cause cancer, disrupt endocrine function, and cause neurotoxicity ^{17 18 60}. These findings are published in detailed technical reports and are stored in public databases like the Chemical Effects in Biological Systems (CEBS), which houses data from over 11,000 test articles ^{20 140143}. The NTP's work provides the fundamental hazard identification data necessary to argue that certain pesticides are inherently dangerous.

Complementing the NTP's mechanistic studies, the EPA's Integrated Risk Information System (IRIS) program compiles and evaluates hazard and dose-response data for thousands of chemicals to assess their potential to cause adverse health effects in humans ^{13 64}. IRIS assessments are authoritative documents that identify and characterize health hazards, often resulting in a reference dose (RfD) or other values intended to be protective of human health ^{12 14}. These assessments are publicly available and serve as the scientific basis for regulatory risk assessment under various environmental laws ⁶⁵. The IRIS database can be searched and downloaded, including weekly data dumps in Excel format, making its contents readily accessible for analysis ^{95 97}. Together, the NTP and IRIS provide a powerful dual-source confirmation: the NTP identifies *what* a chemical can do to biological systems, and the IRIS program attempts to quantify *how much* of that chemical is considered safe.

Perhaps the most persuasive evidence for the public is human biomonitoring data, which measures actual chemical exposures in people. The CDC's NHANES program provides a

continuous survey of the U.S. population, measuring biomarkers of exposure in blood and urine samples ¹. Since 1999, NHANES has been able to detect metabolites of various pesticides in the bodies of a significant portion of Americans ⁵. Studies using NHANES data have found associations between exposure to specific pesticides and adverse health outcomes. For example, research has linked urinary metabolites of non-persistent pesticides like organophosphates and pyrethroids to disruptions in adrenal and gonadal hormones ¹⁰⁴. Other analyses of NHANES data from 2007–2018 have examined associations between five different pesticide biomarkers and increased mortality from cardiovascular disease ¹⁰⁰. Further studies have explored links between pyrethroid exposure and osteoarthritis, and between agricultural pesticide use and Alzheimer's disease dementia ¹⁰¹¹⁰². These epidemiological studies create a direct line from agricultural chemical use to measurable human health impacts, providing the final piece of the evidence chain required for a strong advocacy case.

Synthesizing Data for Advocacy: Frameworks, Gaps, and Strategic Pathways

The ultimate value of the identified datasets lies not in their individual existence but in their synthesis into a coherent, evidence-based narrative. A successful advocacy effort to urge the FDA to ban certain pesticides and mandate regenerative farming must weave together the four pillars of pesticide use, soil health, crop nutrition, and toxicology into a compelling story of cause and effect. This synthesis requires acknowledging inherent data gaps while strategically leveraging the strengths of each dataset to build a robust case.

The most significant insight from this analysis is that the power of the evidence emerges from connecting disparate sources. The argument should proceed along a logical pathway: 1. **Inputs:** Demonstrate the dramatic increase in chemical inputs over the latter half of the 20th century using data from the USGS Pesticide National Synthesis Project and state-level reports like CalPIP ^{81 129}. 2. **Environmental Impact:** Show the consequences of these inputs on the environment using data from the USDA's LTAR and LTER networks, which provide decades of data on soil organic carbon, erosion, and water quality ^{76 87}, and the NRI, which documents national trends in soil erosion ⁵⁶. 3. **Food Quality:** Analyze the impact on food itself, using USDA FoodData Central to explore trends in crop nutritional content, while carefully framing the narrative around "instability" rather than a simplistic claim of universal decline to maintain scientific credibility ^{29 158}. 4. **Human Health:** Provide the final, critical link by using CDC's

NHANES data to prove that humans are being exposed to these pesticides, and by citing NTP and IRIS data to establish their inherent toxicity, creating a direct connection between agricultural practice and public health risk ^{13 17 100}.

Despite the wealth of available data, significant gaps and limitations must be acknowledged. A truly 100-year continuous dataset for any single metric does not exist. The most robust records for pesticide use, soil health, and human exposure generally begin in the 1970s, 1930s, and 1999, respectively. Furthermore, much of the data is fragmented across multiple agency websites and portals, requiring significant effort to integrate. Accessing and curating this information is a major challenge, which is why organizations like the Bionutrient Food Association exist—to try and make sense of this complex landscape ⁸⁴.

For the advocacy effort, the strategic path forward involves several key actions. First, prioritize the creation of unified visualizations and summary metrics that connect the four data pillars. A dashboard showing correlated trends—for example, plotting pesticide use intensity (from PNSP) against soil carbon levels (from LTAR) and nutrient density (from FoodData Central) for a specific region—would be a powerful communication tool. Second, focus the argument on specific, well-studied chemicals like glyphosate or chlorpyrifos, for which the evidence chain from NTP/IRIS to NHANES is strongest. Third, frame the call for regenerative farming not as a vague ideal but as a solution to the documented problems of soil degradation and water pollution, using LTAR data to quantify the benefits of conservation systems. Finally, leverage existing regulatory frameworks. The FDA's authority under the Food Safety Modernization Act (FSMA) recognizes that ensuring food safety is a shared responsibility involving agriculture and the environment ¹⁵³. The agency already assesses risks from agricultural inputs like raw manure, setting a precedent for considering the broader impacts of pesticide use ¹⁵⁵. Collaborative agreements between the FDA, USDA, and EPA further create a procedural pathway for using the synthesized data from these agencies to inform new regulations ¹⁵⁴. By building a meticulously sourced, transparently argued, and scientifically grounded case, this data-driven initiative can move from identifying problems to demanding actionable solutions.

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